```
2177  0  INSERT FILE='E:/Documents/GitHub/biostatistics-portfolio/_spss_tmp/bodyfat-regre
ssion.sps' ERROR=CONTINUE.
2177  0  * Set once: the folder that contains this repository, then run.
2178  0  CD 'E:/Documents/GitHub/biostatistics-portfolio'.
2179  0  GET FILE='data/raw/body fat final.sav'.
2180  0  DATASET NAME DataSet1 WINDOW=FRONT.
```

Dataset Name

Notes

Output Created		03-NOV-2026 15:38:30
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Filter	<none>
	Weight	<none>
	Split File	<none>
Syntax		DATASET NAME DataSet1 ...
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Warnings

The active dataset will replace the existing dataset named DataSet1.

```
2181  0  * Encoding: UTF-8.
2182  0
2183  0  DATASET ACTIVATE DataSet1.
2184  0  DESCRIPTIVES VARIABLES=fat age weight height neck chest abdomen hip thigh knee a
nkle biceps forearm
2185  0      wrist
2186  0    /STATISTICS=MEAN STDDEV MIN MAX.
```

Descriptives

Notes

Output Created		03-NOV-2026 15:38:31
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	252
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=fat age weight height neck chest abdomen hip thigh knee ankle biceps forearm wrist /STATISTICS=MEAN STDDEV MIN MAX.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.02

```
[DataSet1] E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
```

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Body Fat	252	.0	47.5	19.151	8.3687
Age	252	22	81	44.88	12.602
Weight in pounds	252	118.50	363.15	178.9244	29.38916
Height in inches	252	29.50	77.75	70.1488	3.66286
Neck Circumference	252	31.1	51.2	37.992	2.4309
Chest Circumference	252	79.3	136.2	100.824	8.4305
Abdomen Circumference at Umbilicus	252	69.4	148.1	92.556	10.7831
Hip Circumference	252	85.0	147.7	99.905	7.1641
Thigh Circumference	252	47.2	87.3	59.406	5.2500
Knee Circumference	252	33.0	49.1	38.590	2.4118
Ankle Circumference	252	19.1	33.9	23.102	1.6949
Extended Biceps Circumference	252	24.8	45.0	32.273	3.0213
Forearm Circumference	252	21.0	34.9	28.664	2.0207
Wrist Circumference	252	15.8	21.4	18.230	.9336
Valid N (listwise)	252				

```
2187  0
2188  0  *data cleaning
2189  0
2190  0  USE ALL.
2191  0  COMPUTE filter_$=(fat > 0 & height > 30).
2192  0  VARIABLE LABELS filter_$ 'fat > 0 & height > 30 (FILTER)'.
2193  0  VALUE LABELS filter_$ 0 'Not Selected' 1 'Selected'.
2194  0  FORMATS filter_$ (f1.0).
2195  0  FILTER BY filter_$.
2196  0  EXECUTE.
2197  0
2198  0  DESCRIPTIVES VARIABLES=fat age weight height neck chest abdomen hip thigh knee a
nkle biceps forearm
2199  0      wrist
2200  0    /STATISTICS=MEAN STDDEV MIN MAX.
```

Descriptives

Notes

Output Created		03-NOV-2026 15:38:31
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=fat age weight height neck chest abdomen hip thigh knee ankle biceps forearm wrist /STATISTICS=MEAN STDDEV MIN MAX.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.02

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Body Fat	250	.7	47.5	19.172	8.2684
Age	250	22	81	44.91	12.649
Weight in pounds	250	125.00	363.15	179.0618	29.21040
Height in inches	250	64.00	77.75	70.3200	2.61543
Neck Circumference	250	31.1	51.2	38.014	2.4244
Chest Circumference	250	83.4	136.2	100.890	8.3469
Abdomen Circumference at Umbilicus	250	70.4	148.1	92.602	10.7004
Hip Circumference	250	85.3	147.7	99.902	7.0617
Thigh Circumference	250	49.3	87.3	59.410	5.1654
Knee Circumference	250	33.0	49.1	38.595	2.3871
Ankle Circumference	250	19.1	33.9	23.112	1.6913
Extended Biceps Circumference	250	24.8	45.0	32.286	3.0183
Forearm Circumference	250	21.0	34.9	28.680	2.0123
Wrist Circumference	250	15.8	21.4	18.240	.9294
Valid N (listwise)	250				

```
2201  0
2202  0  *fit model; check all 5 assumptions; collinearity and outliers
2203  0
2204  0  REGRESSION
2205  0    /MISSING LISTWISE
2206  0    /STATISTICS COEFF OUTS R ANOVA COLLIN TOL ZPP
2207  0    /CRITERIA=PIN(.05) POUT(.10)
2208  0    /NOORIGIN
2209  0    /DEPENDENT fat
2210  0    /METHOD=ENTER age weight height neck chest abdomen hip thigh knee ankle biceps
 forearm wrist
2211  0    /SCATTERPLOT=(*ZRESID ,*ZPRED)
2212  0    /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID)
2213  0    /CASEWISE PLOT(ZRESID) OUTLIERS(3)
2214  0    /SAVE PRED COOK RESID SDBETA.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:31
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA COLLIN TOL ZPP /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age weight height neck chest abdomen hip thigh knee ankle biceps forearm wrist /SCATTERPLOT=(*ZRESID ,*ZPRED) /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID) /CASEWISE PLOT(ZRESID) OUTLIERS(3) /SAVE PRED COOK RESID SDBETA.
Resources	Processor Time	00:00:01.25
	Elapsed Time	00:00:01.53

Notes

	Memory Required	14352 bytes
	Additional Memory Required for Residual Plots	704 bytes
Variables Created or Modified	PRE_3	Unstandardized Predicted Value
	RES_3	Unstandardized Residual
	COO_3	Cook's Distance
	SDB0_3	Standardized DFBETA for (Constant)
	SDB1_3	Standardized DFBETA for age
	SDB2_3	Standardized DFBETA for weight
	SDB3_3	Standardized DFBETA for height
	SDB4_3	Standardized DFBETA for neck
	SDB5_3	Standardized DFBETA for chest
	SDB6_3	Standardized DFBETA for abdomen
	SDB7_3	Standardized DFBETA for hip
	SDB8_3	Standardized DFBETA for thigh
	SDB9_3	Standardized DFBETA for knee
	SDB10_3	Standardized DFBETA for ankle
	SDB11_3	Standardized DFBETA for biceps
	SDB12_3	Standardized DFBETA for forearm
	SDB13_3	Standardized DFBETA for wrist

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Abdomen Circumference at Umbilicus, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Thigh Circumference, Chest Circumference, Hip Circumference, Weight in pounds[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.861[a]	.742	.728	4.3130	1.785

a. Predictors: (Constant), Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Abdomen Circumference at Umbilicus, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Thigh Circumference, Chest Circumference, Hip Circumference, Weight in pounds

b. Dependent Variable: Body Fat

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	12633.110	13	971.778	52.242	.000[b]
	Residual	4389.969	236	18.602		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Abdomen Circumference at Umbilicus, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Thigh Circumference, Chest Circumference, Hip Circumference, Weight in pounds

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-17.163	22.458		-.764	.446
	Age	.061	.032	.093	1.883	.061
	Weight in pounds	-.087	.062	-.306	-1.387	.167
	Height in inches	-.061	.179	-.019	-.339	.735
	Neck Circumference	-.462	.236	-.135	-1.954	.052
	Chest Circumference	-.034	.104	-.035	-.331	.741
	Abdomen Circumference at Umbilicus	.959	.090	1.242	10.609	.000
	Hip Circumference	-.209	.147	-.179	-1.424	.156
	Thigh Circumference	.222	.148	.139	1.506	.133
	Knee Circumference	.007	.248	.002	.028	.978
	Ankle Circumference	.167	.223	.034	.747	.456
	Extended Biceps Circumference	.195	.173	.071	1.128	.260
	Forearm Circumference	.439	.200	.107	2.197	.029
	Wrist Circumference	-1.632	.536	-.183	-3.044	.003

Coefficients[a]

Model		Correlations			Collinearity Statistics	
		Zero-order	Partial	Part	Tolerance	VIF
1	(Constant)					
	Age	.293	.122	.062	.444	2.251
	Weight in pounds	.603	-.090	-.046	.022	44.449
	Height in inches	-.032	-.022	-.011	.339	2.950
	Neck Circumference	.490	-.126	-.065	.228	4.395
	Chest Circumference	.696	-.022	-.011	.098	10.156
	Abdomen Circumference at Umbilicus	.809	.568	.351	.080	12.533
	Hip Circumference	.613	-.092	-.047	.069	14.402
	Thigh Circumference	.544	.098	.050	.128	7.791
	Knee Circumference	.494	.002	.001	.213	4.698
	Ankle Circumference	.254	.049	.025	.525	1.905
	Extended Biceps Circumference	.487	.073	.037	.274	3.647
	Forearm Circumference	.351	.142	.073	.462	2.166
	Wrist Circumference	.344	-.194	-.101	.301	3.324

a. Dependent Variable: Body Fat

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Constant)	Age	Weight in pounds
1	1	13.891	1.000	.00	.00	.00
	2	.074	13.737	.00	.40	.00
	3	.020	26.434	.00	.01	.01
	4	.004	60.110	.00	.00	.00
	5	.003	66.712	.00	.06	.02
	6	.002	81.369	.00	.06	.01
	7	.002	86.503	.00	.04	.06
	8	.001	96.781	.00	.10	.00
	9	.001	111.421	.00	.00	.00
	10	.001	146.754	.00	.17	.02
	11	.001	148.162	.00	.02	.00
	12	.001	155.027	.00	.14	.02
	13	.000	194.015	.00	.00	.00
	14	8.990E-5	393.076	.98	.00	.87

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions				
		Height in inches	Neck Circumference	Chest Circumference	Abdomen Circumference at Umbilicus	Hip Circumference
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.00	.00	.00
	3	.00	.00	.00	.01	.00
	4	.00	.00	.00	.00	.00
	5	.00	.00	.01	.06	.01
	6	.00	.01	.02	.01	.00
	7	.06	.00	.03	.08	.00
	8	.01	.08	.04	.00	.00
	9	.06	.36	.03	.01	.00
	10	.10	.06	.09	.01	.01
	11	.00	.21	.15	.12	.04
	12	.00	.13	.35	.50	.01
	13	.15	.02	.01	.17	.57
	14	.61	.11	.28	.02	.35

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions				
		Thigh Circumference	Knee Circumference	Ankle Circumference	Extended Biceps Circumference	Forearm Circumference
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.00	.00	.00
	3	.00	.00	.00	.00	.00
	4	.00	.01	.18	.22	.16
	5	.01	.00	.35	.05	.02
	6	.10	.01	.00	.33	.28
	7	.00	.03	.34	.02	.01
	8	.13	.05	.00	.15	.46
	9	.13	.02	.00	.13	.01
	10	.01	.39	.08	.00	.05
	11	.00	.24	.01	.01	.00
	12	.20	.21	.00	.06	.00
	13	.39	.03	.00	.00	.01
	14	.02	.00	.04	.02	.00

Collinearity Diagnostics[a]

Model	Dimension	Variance ... Wrist Circumference
1	1	.00
	2	.00
	3	.00
	4	.00
	5	.00
	6	.01
	7	.01
	8	.02
	9	.06
	10	.39
	11	.46
	12	.01
	13	.03
	14	.01

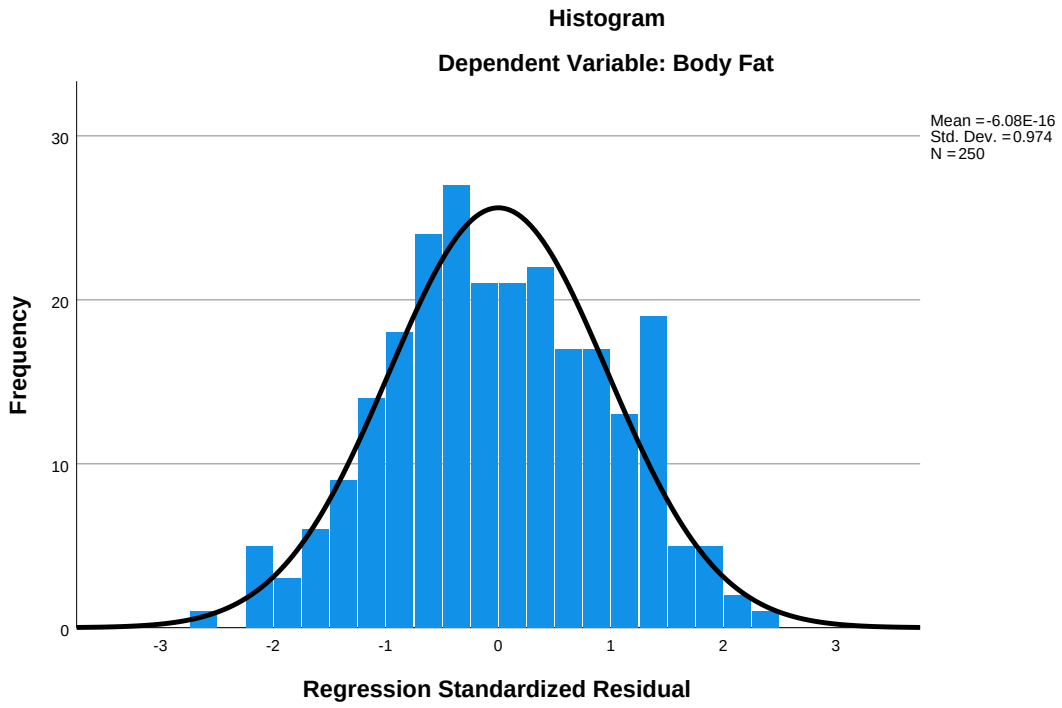
a. Dependent Variable: Body Fat

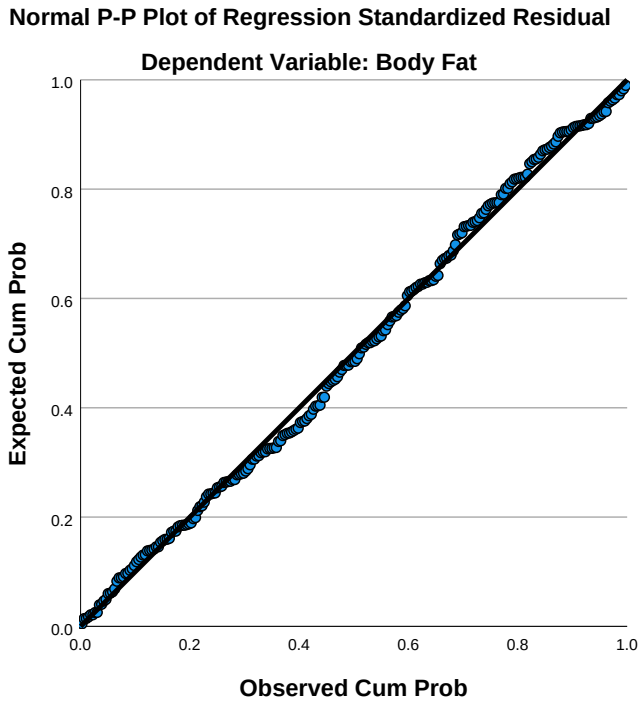
Residuals Statistics[a]

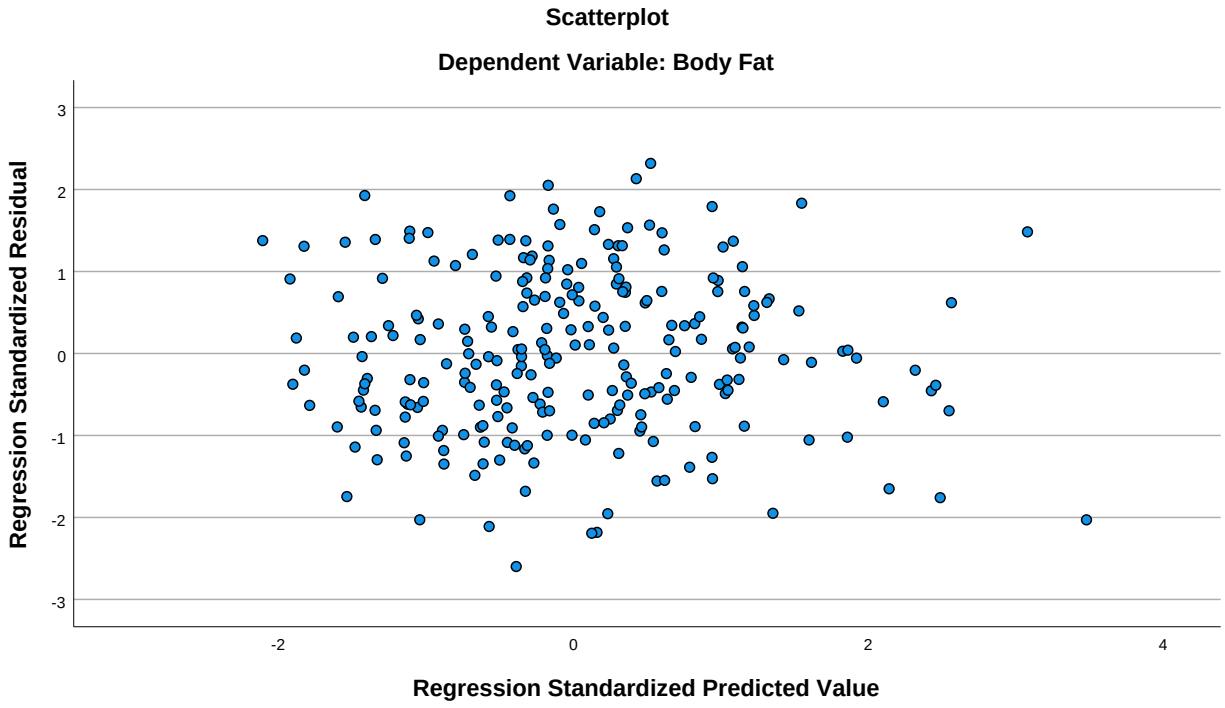
	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	4.162	43.952	19.172	7.1229	250
Std. Predicted Value	-2.107	3.479	.000	1.000	250
Standard Error of Predicted Value	.493	2.919	.971	.316	250
Adjusted Predicted Value	3.933	51.346	19.187	7.2272	250
Residual	-11.2082	9.9999	.0000	4.1989	250
Std. Residual	-2.599	2.319	.000	.974	250
Stud. Residual	-2.756	2.374	-.001	1.007	250
Deleted Residual	-16.1458	10.4832	-.0149	4.5201	250
Stud. Deleted Residual	-2.796	2.398	-.001	1.010	250
Mahal. Distance	2.255	113.030	12.948	12.168	250
Cook's Distance	.000	.458	.006	.030	250
Centered Leverage Value	.009	.454	.052	.049	250

a. Dependent Variable: Body Fat

Charts


Histogram
Dependent Variable: Body Fat
Mean =-6.08E-16
Std. Dev. =0.974
N =250
Frequency
30
20
10
0
-3
-2
-1
0
1
2
3
Regression Standardized Residual


Normal P-P Plot of Regression Standardized Residual
Dependent Variable: Body Fat
Expected Cum Prob
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Observed Cum Prob




```
2215  0
2216  0  DATASET ACTIVATE DataSet1.
2217  0  REGRESSION
2218  0    /MISSING LISTWISE
2219  0    /STATISTICS COEFF OUTS R ANOVA
2220  0    /CRITERIA=PIN(.05) POUT(.10)
2221  0    /NOORIGIN
2222  0    /DEPENDENT RES_1
2223  0    /METHOD=ENTER PRE_1.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:33
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT RES_1 /METHOD=ENTER PRE_1.
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.05
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Unstandardized Predicted Value[b]	.	Enter

a. Dependent Variable: Unstandardized Residual

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.000[a]	.000	-.004	4.20731380

a. Predictors: (Constant), Unstandardized Predicted Value

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	.000	1	.000	.000	1.000[b]
	Residual	4389.969	248	17.701		
	Total	4389.969	249			

a. Dependent Variable: Unstandardized Residual

b. Predictors: (Constant), Unstandardized Predicted Value

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-2.615E-15	.765		.000	1.000
	Unstandardized Predicted Value	.000	.037	.000	.000	1.000

a. Dependent Variable: Unstandardized Residual

```
2224  0
2225  0  *correlate the predictors to check which correlates with weight the most
2226  0
2227  0
```

```
2228  0  NONPAR CORR
2229  0    /VARIABLES=age weight height neck chest abdomen hip thigh knee ankle biceps fo
rearm wrist
2230  0    /PRINT=SPEARMAN TWOTAIL NOSIG FULL
2231  0    /MISSING=PAIRWISE.
```

Nonparametric Correlations

Notes

Output Created		03-NOV-2026 15:38:33
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each pair of variables are based on all the cases with valid data for that pair.
Syntax		NONPAR CORR /VARIABLES=age weight height neck chest abdomen hip thigh knee ankle biceps forearm wrist /PRINT=SPEARMAN TWOTAIL NOSIG FULL /MISSING=PAIRWISE.
Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.04
	Number of Cases Allowed	196608 cases[a]

a. Based on availability of workspace memory

Correlations

			Age	Weight in pounds
Spearman's rho	Age	Correlation Coefficient	1.000	-.017
		Sig. (2-tailed)	.	.785
		N	250	250
	Weight in pounds	Correlation Coefficient	-.017	1.000
		Sig. (2-tailed)	.785	.
		N	250	250
	Height in inches	Correlation Coefficient	-.234**	.525**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Neck Circumference	Correlation Coefficient	.120	.808**
		Sig. (2-tailed)	.057	.000
		N	250	250
	Chest Circumference	Correlation Coefficient	.165**	.896**
		Sig. (2-tailed)	.009	.000
		N	250	250
	Abdomen Circumference at Umbilicus	Correlation Coefficient	.219**	.872**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Hip Circumference	Correlation Coefficient	-.078	.928**
		Sig. (2-tailed)	.221	.000
		N	250	250
	Thigh Circumference	Correlation Coefficient	-.210**	.834**
		Sig. (2-tailed)	.001	.000
		N	250	250
	Knee Circumference	Correlation Coefficient	.001	.826**
		Sig. (2-tailed)	.983	.000
		N	250	250
	Ankle Circumference	Correlation Coefficient	-.136*	.691**
		Sig. (2-tailed)	.032	.000
		N	250	250

Correlations

			Height in inches	Neck Circumference
Spearman's rho	Age	Correlation Coefficient	-.234**	.120
		Sig. (2-tailed)	.000	.057
		N	250	250
	Weight in pounds	Correlation Coefficient	.525**	.808**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Height in inches	Correlation Coefficient	1.000	.314**
		Sig. (2-tailed)	.	.000
		N	250	250
	Neck Circumference	Correlation Coefficient	.314**	1.000
		Sig. (2-tailed)	.000	.
		N	250	250
	Chest Circumference	Correlation Coefficient	.261**	.784**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Abdomen Circumference at Umbilicus	Correlation Coefficient	.225**	.747**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Hip Circumference	Correlation Coefficient	.434**	.715**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Thigh Circumference	Correlation Coefficient	.344**	.654**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Knee Circumference	Correlation Coefficient	.507**	.644**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Ankle Circumference	Correlation Coefficient	.463**	.515**
		Sig. (2-tailed)	.000	.000
		N	250	250

Correlations

			Chest Circumference	Abdomen Circumference at Umbilicus
Spearman's rho	Age	Correlation Coefficient	.165**	.219**
		Sig. (2-tailed)	.009	.000
		N	250	250
	Weight in pounds	Correlation Coefficient	.896**	.872**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Height in inches	Correlation Coefficient	.261**	.225**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Neck Circumference	Correlation Coefficient	.784**	.747**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Chest Circumference	Correlation Coefficient	1.000	.894**
		Sig. (2-tailed)	.	.000
		N	250	250
	Abdomen Circumference at Umbilicus	Correlation Coefficient	.894**	1.000
		Sig. (2-tailed)	.000	.
		N	250	250
	Hip Circumference	Correlation Coefficient	.809**	.842**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Thigh Circumference	Correlation Coefficient	.716**	.729**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Knee Circumference	Correlation Coefficient	.706**	.710**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Ankle Circumference	Correlation Coefficient	.553**	.500**
		Sig. (2-tailed)	.000	.000
		N	250	250

Correlations

			Hip Circumference	Thigh Circumference
Spearman's rho	Age	Correlation Coefficient	-.078	-.210**
		Sig. (2-tailed)	.221	.001
		N	250	250
	Weight in pounds	Correlation Coefficient	.928**	.834**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Height in inches	Correlation Coefficient	.434**	.344**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Neck Circumference	Correlation Coefficient	.715**	.654**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Chest Circumference	Correlation Coefficient	.809**	.716**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Abdomen Circumference at Umbilicus	Correlation Coefficient	.842**	.729**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Hip Circumference	Correlation Coefficient	1.000	.875**
		Sig. (2-tailed)	.	.000
		N	250	250
	Thigh Circumference	Correlation Coefficient	.875**	1.000
		Sig. (2-tailed)	.000	.
		N	250	250
	Knee Circumference	Correlation Coefficient	.798**	.765**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Ankle Circumference	Correlation Coefficient	.620**	.607**
		Sig. (2-tailed)	.000	.000
		N	250	250

Correlations

			Knee Circumference	Ankle Circumference
Spearman's rho	Age	Correlation Coefficient	.001	-.136*
		Sig. (2-tailed)	.983	.032
		N	250	250
	Weight in pounds	Correlation Coefficient	.826**	.691**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Height in inches	Correlation Coefficient	.507**	.463**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Neck Circumference	Correlation Coefficient	.644**	.515**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Chest Circumference	Correlation Coefficient	.706**	.553**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Abdomen Circumference at Umbilicus	Correlation Coefficient	.710**	.500**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Hip Circumference	Correlation Coefficient	.798**	.620**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Thigh Circumference	Correlation Coefficient	.765**	.607**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Knee Circumference	Correlation Coefficient	1.000	.722**
		Sig. (2-tailed)	.	.000
		N	250	250
	Ankle Circumference	Correlation Coefficient	.722**	1.000
		Sig. (2-tailed)	.000	.
		N	250	250

Correlations

			Extended Biceps Circumference	Forearm Circumference
Spearman's rho	Age	Correlation Coefficient	-.049	-.072
		Sig. (2-tailed)	.445	.257
		N	250	250
	Weight in pounds	Correlation Coefficient	.779**	.749**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Height in inches	Correlation Coefficient	.304**	.332**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Neck Circumference	Correlation Coefficient	.695**	.711**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Chest Circumference	Correlation Coefficient	.736**	.679**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Abdomen Circumference at Umbilicus	Correlation Coefficient	.669**	.592**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Hip Circumference	Correlation Coefficient	.734**	.680**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Thigh Circumference	Correlation Coefficient	.734**	.665**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Knee Circumference	Correlation Coefficient	.627**	.624**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Ankle Circumference	Correlation Coefficient	.521**	.558**
		Sig. (2-tailed)	.000	.000
		N	250	250

Correlations

			Wrist Circumference
Spearman's rho	Age	Correlation Coefficient	.215**
		Sig. (2-tailed)	.001
		N	250
	Weight in pounds	Correlation Coefficient	.703**
		Sig. (2-tailed)	.000
		N	250
	Height in inches	Correlation Coefficient	.390**
		Sig. (2-tailed)	.000
		N	250
	Neck Circumference	Correlation Coefficient	.718**
		Sig. (2-tailed)	.000
		N	250
	Chest Circumference	Correlation Coefficient	.653**
		Sig. (2-tailed)	.000
		N	250
	Abdomen Circumference at Umbilicus	Correlation Coefficient	.598**
		Sig. (2-tailed)	.000
		N	250
	Hip Circumference	Correlation Coefficient	.599**
		Sig. (2-tailed)	.000
		N	250
	Thigh Circumference	Correlation Coefficient	.501**
		Sig. (2-tailed)	.000
		N	250
	Knee Circumference	Correlation Coefficient	.654**
		Sig. (2-tailed)	.000
		N	250
	Ankle Circumference	Correlation Coefficient	.653**
		Sig. (2-tailed)	.000
		N	250

Correlations

		Age	Weight in pounds
Extended Biceps Circumference	Correlation Coefficient	-.049	.779**
	Sig. (2-tailed)	.445	.000
	N	250	250
Forearm Circumference	Correlation Coefficient	-.072	.749**
	Sig. (2-tailed)	.257	.000
	N	250	250
Wrist Circumference	Correlation Coefficient	.215**	.703**
	Sig. (2-tailed)	.001	.000
	N	250	250

Correlations

		Height in inches	Neck Circumference
Extended Biceps Circumference	Correlation Coefficient	.304**	.695**
	Sig. (2-tailed)	.000	.000
	N	250	250
Forearm Circumference	Correlation Coefficient	.332**	.711**
	Sig. (2-tailed)	.000	.000
	N	250	250
Wrist Circumference	Correlation Coefficient	.390**	.718**
	Sig. (2-tailed)	.000	.000
	N	250	250

Correlations

			Chest Circumference	Abdomen Circumference at Umbilicus
	Extended Biceps Circumference	Correlation Coefficient	.736**	.669**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Forearm Circumference	Correlation Coefficient	.679**	.592**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Wrist Circumference	Correlation Coefficient	.653**	.598**
		Sig. (2-tailed)	.000	.000
		N	250	250

Correlations

			Hip Circumference	Thigh Circumference
	Extended Biceps Circumference	Correlation Coefficient	.734**	.734**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Forearm Circumference	Correlation Coefficient	.680**	.665**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Wrist Circumference	Correlation Coefficient	.599**	.501**
		Sig. (2-tailed)	.000	.000
		N	250	250

Correlations

			Knee Circumference	Ankle Circumference
	Extended Biceps Circumference	Correlation Coefficient	.627**	.521**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Forearm Circumference	Correlation Coefficient	.624**	.558**
		Sig. (2-tailed)	.000	.000
		N	250	250
	Wrist Circumference	Correlation Coefficient	.654**	.653**
		Sig. (2-tailed)	.000	.000
		N	250	250

Correlations

			Extended Biceps Circumference	Forearm Circumference
	Extended Biceps Circumference	Correlation Coefficient	1.000	.762**
		Sig. (2-tailed)	.	.000
		N	250	250
	Forearm Circumference	Correlation Coefficient	.762**	1.000
		Sig. (2-tailed)	.000	.
		N	250	250
	Wrist Circumference	Correlation Coefficient	.602**	.639**
		Sig. (2-tailed)	.000	.000
		N	250	250

Correlations

			Wrist Circumference
	Extended Biceps Circumference	Correlation Coefficient	.602**
		Sig. (2-tailed)	.000
		N	250
	Forearm Circumference	Correlation Coefficient	.639**
		Sig. (2-tailed)	.000
		N	250
	Wrist Circumference	Correlation Coefficient	1.000
		Sig. (2-tailed)	.
		N	250

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

```
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2233  0  *regress again to look at vif when weight is removed
2234  0
2235  0
2236  0
2237  0  REGRESSION
2238  0    /MISSING LISTWISE
2239  0    /STATISTICS R ANOVA COLLIN TOL
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2241  0    /NOORIGIN
2242  0    /DEPENDENT fat
2243  0    /METHOD=ENTER age height neck chest abdomen hip thigh knee ankle biceps forear
m wrist
2244  0    /SCATTERPLOT=(*ZRESID ,*ZPRED).
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Regression

Notes

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	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS R ANOVA COLLIN TOL /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age height neck chest abdomen hip thigh knee ankle biceps forearm wrist /SCATTERPLOT=(*ZRESID ,*ZPRED).
Resources	Processor Time	00:00:00.64
	Elapsed Time	00:00:00.63
	Memory Required	13872 bytes
	Additional Memory Required for Residual Plots	48 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Abdomen Circumference at Umbilicus, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Thigh Circumference, Chest Circumference, Hip Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.860[a]	.740	.727	4.3213

a. Predictors: (Constant), Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Abdomen Circumference at Umbilicus, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Thigh Circumference, Chest Circumference, Hip Circumference

b. Dependent Variable: Body Fat

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	12597.333	12	1049.778	56.216	.000[b]
	Residual	4425.746	237	18.674		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Abdomen Circumference at Umbilicus, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Thigh Circumference, Chest Circumference, Hip Circumference

Coefficients[a]

Model		Collinearity Statistics	
		Tolerance	VIF
1	Age	.457	2.186
	Height in inches	.571	1.750
	Neck Circumference	.256	3.912
	Chest Circumference	.129	7.760
	Abdomen Circumference at Umbilicus	.088	11.328
	Hip Circumference	.094	10.613
	Thigh Circumference	.131	7.630
	Knee Circumference	.215	4.644
	Ankle Circumference	.549	1.823
	Extended Biceps Circumference	.286	3.495
	Forearm Circumference	.462	2.166
	Wrist Circumference	.307	3.262

a. Dependent Variable: Body Fat

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Variance Proportions Age	Variance Proportions Height in inches
1	1	12.903	1.000	.00	.00	.00
	2	.072	13.430	.00	.43	.00
	3	.011	33.712	.02	.00	.01
	4	.004	58.329	.00	.01	.00
	5	.003	67.887	.04	.03	.02
	6	.002	78.968	.00	.09	.01
	7	.001	93.195	.02	.11	.01
	8	.001	106.505	.04	.00	.05
	9	.001	119.607	.29	.04	.07
	10	.001	142.764	.00	.03	.00
	11	.001	145.140	.01	.00	.17
	12	.000	173.993	.58	.24	.55
	13	.000	187.721	.00	.01	.11

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions Neck Circumference	Variance Proportions Chest Circumference	Variance Proportions Abdomen Circumference at Umbilicus	Variance Proportions Hip Circumference	Variance Proportions Thigh Circumference
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.00	.00	.00
	3	.00	.01	.04	.00	.00
	4	.00	.00	.02	.00	.00
	5	.00	.00	.01	.00	.00
	6	.01	.05	.03	.00	.09
	7	.08	.06	.00	.00	.12
	8	.44	.05	.03	.00	.08
	9	.00	.00	.00	.00	.16
	10	.29	.12	.10	.04	.00
	11	.14	.54	.34	.01	.06
	12	.01	.17	.16	.03	.16
	13	.03	.00	.27	.90	.33

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Knee Circumference	Ankle Circumference	Extended Biceps Circumference	Forearm Circumference	Wrist Circumference
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.00	.00	.00
	3	.00	.01	.01	.00	.00
	4	.01	.12	.23	.18	.00
	5	.00	.65	.09	.00	.00
	6	.02	.05	.25	.29	.00
	7	.06	.00	.18	.46	.01
	8	.00	.01	.16	.01	.10
	9	.34	.09	.00	.01	.07
	10	.15	.02	.01	.00	.62
	11	.05	.04	.02	.03	.15
	12	.36	.00	.04	.01	.00
	13	.01	.00	.00	.01	.04

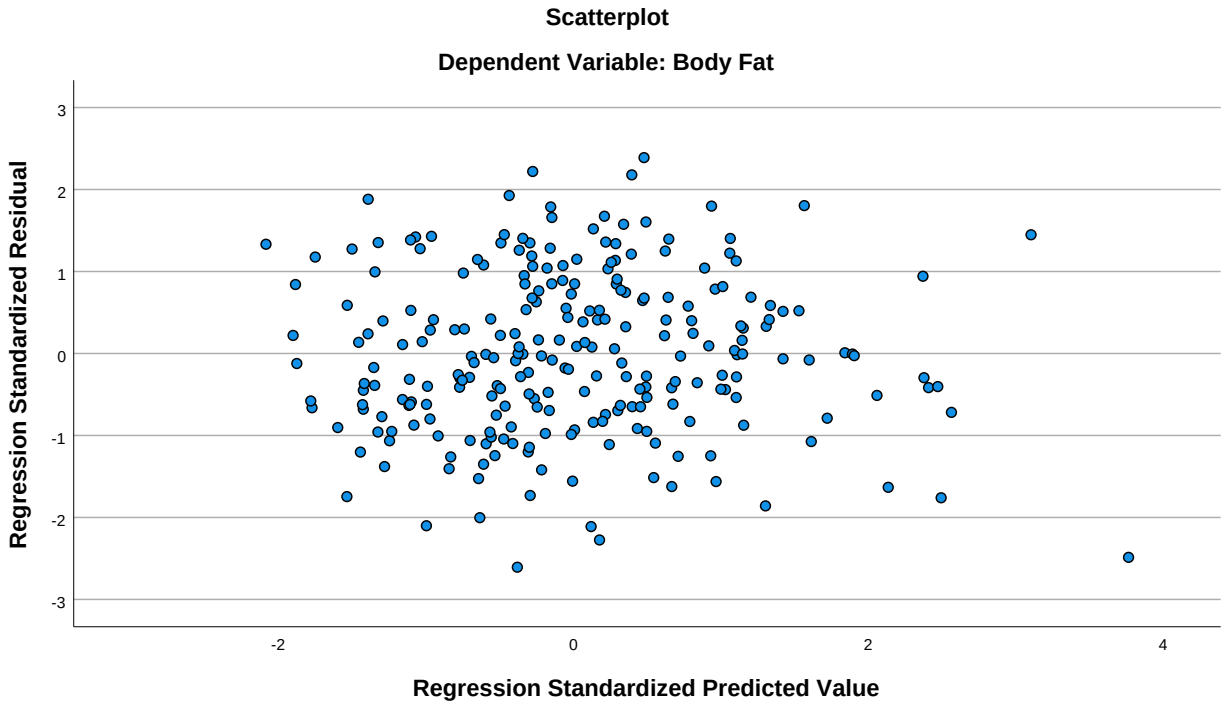
a. Dependent Variable: Body Fat

Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	4.343	45.946	19.172	7.1128	250
Residual	-11.2660	10.3239	.0000	4.2159	250
Std. Predicted Value	-2.085	3.764	.000	1.000	250
Std. Residual	-2.607	2.389	.000	.976	250

a. Dependent Variable: Body Fat

Charts




```
2245  0
2246  0  *regress again to look at vif when abdominal circumference at umbilicus is remov
ed
2247  0
2248  0  REGRESSION
2249  0    /MISSING LISTWISE
2250  0    /STATISTICS R ANOVA COLLIN TOL
2251  0    /CRITERIA=PIN(.05) POUT(.10)
2252  0    /NOORIGIN
2253  0    /DEPENDENT fat
2254  0    /METHOD=ENTER age height neck chest hip thigh knee ankle biceps forearm wrist
2255  0    /SCATTERPLOT=(*ZRESID ,*ZPRED).
```

Regression

Notes

Output Created		03-NOV-2026 15:38:33
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
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	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS R ANOVA COLLIN TOL /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age height neck chest hip thigh knee ankle biceps forearm wrist /SCATTERPLOT=(*ZRESID ,*ZPRED).
Resources	Processor Time	00:00:00.62
	Elapsed Time	00:00:00.88
	Memory Required	12768 bytes
	Additional Memory Required for Residual Plots	64 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Neck Circumference, Knee Circumference, Hip Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.784[a]	.615	.597	5.2493

a. Predictors: (Constant), Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Neck Circumference, Knee Circumference, Hip Circumference

b. Dependent Variable: Body Fat

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10464.876	11	951.352	34.525	.000[b]
	Residual	6558.204	238	27.555		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Neck Circumference, Knee Circumference, Hip Circumference

Coefficients[a]

Model		Collinearity Statistics	
		Tolerance	VIF
1	Age	.545	1.834
	Height in inches	.582	1.717
	Neck Circumference	.258	3.882
	Chest Circumference	.202	4.955
	Hip Circumference	.118	8.465
	Thigh Circumference	.133	7.496
	Knee Circumference	.216	4.639
	Ankle Circumference	.550	1.817
	Extended Biceps Circumference	.288	3.466
	Forearm Circumference	.463	2.162
	Wrist Circumference	.311	3.221

a. Dependent Variable: Body Fat

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Variance Proportions Age	Variance Proportions Height in inches
1	1	11.908	1.000	.00	.00	.00
	2	.072	12.903	.00	.51	.00
	3	.007	40.010	.03	.01	.02
	4	.003	58.873	.00	.00	.00
	5	.003	66.652	.03	.01	.02
	6	.002	80.990	.01	.06	.02
	7	.001	89.588	.01	.14	.00
	8	.001	104.448	.04	.00	.02
	9	.001	115.031	.32	.05	.07
	10	.001	137.670	.00	.04	.04
	11	.000	157.543	.38	.12	.40
	12	.000	173.889	.16	.06	.40

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions Neck Circumference	Variance Proportions Chest Circumference	Variance Proportions Hip Circumference	Variance Proportions Thigh Circumference	Variance Proportions Knee Circumference
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.00	.00	.00
	3	.00	.02	.00	.01	.00
	4	.00	.00	.01	.02	.01
	5	.00	.01	.02	.02	.00
	6	.03	.23	.00	.02	.01
	7	.10	.12	.00	.12	.06
	8	.40	.30	.00	.06	.00
	9	.00	.01	.00	.16	.36
	10	.43	.00	.03	.00	.06
	11	.00	.07	.38	.02	.37
	12	.03	.25	.54	.57	.12

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Ankle Circumference	Extended Biceps Circumference	Forearm Circumference	Wrist Circumference
1	1	.00	.00	.00	.00
	2	.00	.00	.00	.00
	3	.02	.06	.00	.00
	4	.23	.09	.25	.00
	5	.58	.08	.06	.00
	6	.01	.52	.18	.00
	7	.00	.14	.44	.02
	8	.01	.07	.01	.12
	9	.09	.00	.01	.06
	10	.05	.00	.00	.77
	11	.01	.00	.04	.02
	12	.00	.03	.00	.01

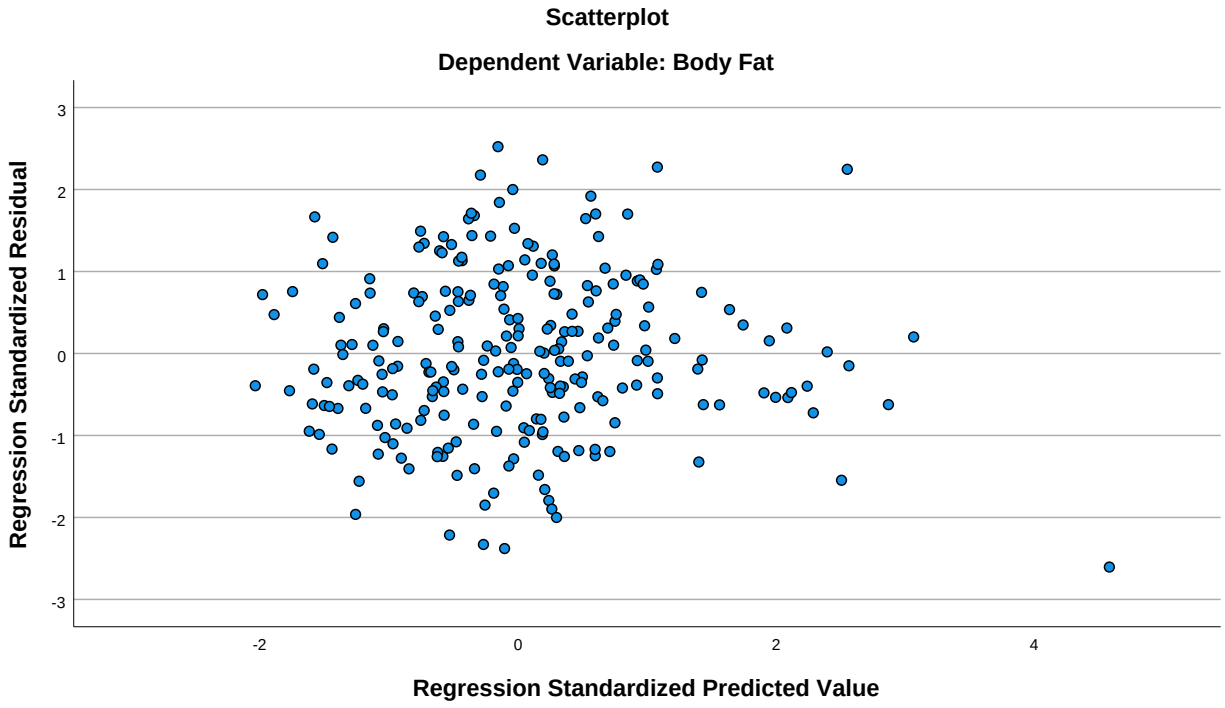
a. Dependent Variable: Body Fat

Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	5.966	48.875	19.172	6.4829	250
Residual	-13.6749	13.2411	.0000	5.1321	250
Std. Predicted Value	-2.037	4.582	.000	1.000	250
Std. Residual	-2.605	2.522	.000	.978	250

a. Dependent Variable: Body Fat

Charts




```
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2258  0  *summarises cooksd and all dfebetas
2259  0
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7_1 SDB8_1 SDB9_1
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Frequencies

Notes

Output Created		03-NOV-2026 15:38:34
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=COO_1 SDB0_1 SDB1_1 SDB2_1 SDB3_1 SDB4_1 SDB5_1 SDB6_1 SDB7_1 SDB8_1 SDB9_1 SDB10_1 SDB11_1 SDB12_1 SDB13_1 /FORMAT=NOTABLE /STATISTICS=MINIMUM MAXIMUM /ORDER=ANALYSIS.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.03

Statistics

		Cook's Distance	Standardized DFBETA Intercept	Standardized DFBETA age	Standardized DFBETA weight	Standardized DFBETA height
N	Valid	250	250	250	250	250
	Missing	0	0	0	0	0
Minimum		.00000	-1.03138	-.30692	-1.26576	-.42852
Maximum		.45840	.59638	.32367	.62318	1.16900

Statistics

		Standardized DFBETA neck	Standardized DFBETA chest	Standardized DFBETA abdomen	Standardized DFBETA hip	Standardized DFBETA thigh
N	Valid	250	250	250	250	250
	Missing	0	0	0	0	0
Minimum		-.33690	-.22857	-.29237	-.32301	-.20230
Maximum		.29136	1.00184	.62137	.25616	.48049

Statistics

		Standardized DFBETA knee	Standardized DFBETA ankle	Standardized DFBETA biceps	Standardized DFBETA forearm	Standardized DFBETA wrist
N	Valid	250	250	250	250	250
	Missing	0	0	0	0	0
Minimum		-.24378	-.50365	-.47861	-.58496	-.30543
Maximum		.21671	1.17294	.25118	.90752	.35382

```
2265  0
2266  0  *manual check for equal variance
2267  0
2268  0  REGRESSION
2269  0    /MISSING LISTWISE
2270  0    /STATISTICS COEFF OUTS R ANOVA
2271  0    /CRITERIA=PIN(.05) POUT(.10)
2272  0    /NOORIGIN
2273  0    /DEPENDENT RES_1
2274  0    /METHOD=ENTER PRE_1.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:34
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT RES_1 /METHOD=ENTER PRE_1.
Resources	Processor Time	00:00:00.11
	Elapsed Time	00:00:00.06
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Unstandardized Predicted Value[b]	.	Enter

a. Dependent Variable: Unstandardized Residual

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.000[a]	.000	-.004	4.20731380

a. Predictors: (Constant), Unstandardized Predicted Value

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	.000	1	.000	.000	1.000[b]
	Residual	4389.969	248	17.701		
	Total	4389.969	249			

a. Dependent Variable: Unstandardized Residual

b. Predictors: (Constant), Unstandardized Predicted Value

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-2.615E-15	.765		.000	1.000
	Unstandardized Predicted Value	.000	.037	.000	.000	1.000

a. Dependent Variable: Unstandardized Residual

```
2275  0
2276  0
2277  0  *Model we are using at this point
2278  0
```

```
2279  0  REGRESSION
2280  0    /MISSING LISTWISE
2281  0    /STATISTICS COEFF OUTS R ANOVA
2282  0    /CRITERIA=PIN(.05) POUT(.10)
2283  0    /NOORIGIN
2284  0    /DEPENDENT fat
2285  0    /METHOD=ENTER age height neck chest hip thigh knee ankle biceps forearm wrist.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:35
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age height neck chest hip thigh knee ankle biceps forearm wrist.
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.06

Notes

Memory Required	12688 bytes
Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Neck Circumference, Knee Circumference, Hip Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.784[a]	.615	.597	5.2493

a. Predictors: (Constant), Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Neck Circumference, Knee Circumference, Hip Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10464.876	11	951.352	34.525	.000[b]
	Residual	6558.204	238	27.555		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Wrist Circumference, Age, Height in inches, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Neck Circumference, Knee Circumference, Hip Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-16.098	10.190		-1.580	.115
	Age	.206	.036	.315	5.785	.000
	Height in inches	-.422	.167	-.133	-2.530	.012
	Neck Circumference	-.362	.270	-.106	-1.337	.182
	Chest Circumference	.482	.089	.487	5.438	.000
	Hip Circumference	.294	.137	.251	2.142	.033
	Thigh Circumference	.401	.176	.250	2.272	.024
	Knee Circumference	.055	.300	.016	.185	.854
	Ankle Circumference	-.023	.265	-.005	-.088	.930
	Extended Biceps Circumference	-.017	.205	-.006	-.084	.933
	Forearm Circumference	.345	.243	.084	1.419	.157
	Wrist Circumference	-2.375	.642	-.267	-3.697	.000

a. Dependent Variable: Body Fat

```
2286  0
2287  0  * CHECK FOR INTERACTION
2288  0      *first run bivariate analysis for each predictor with body fat; we use a cut
t off for 0.2
2289  0
2290  0
2291  0
2292  0  REGRESSION
2293  0    /MISSING LISTWISE
```

```
2294  0    /STATISTICS COEFF OUTS R ANOVA
2295  0    /CRITERIA=PIN(.05) POUT(.10)
2296  0    /NOORIGIN
2297  0    /DEPENDENT fat
2298  0    /METHOD=ENTER age.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:35
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age.
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.08
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Age[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.293[a]	.086	.082	7.9210

a. Predictors: (Constant), Age

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1462.825	1	1462.825	23.315	.000[b]
	Residual	15560.255	248	62.743		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Age

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	10.567	1.851		5.708	.000
	Age	.192	.040	.293	4.829	.000

a. Dependent Variable: Body Fat

```
2299  0
2300  0  REGRESSION
2301  0    /MISSING LISTWISE
2302  0    /STATISTICS COEFF OUTS R ANOVA
2303  0    /CRITERIA=PIN(.05) POUT(.10)
2304  0    /NOORIGIN
2305  0    /DEPENDENT fat
```

```
2306  0    /METHOD=ENTER height.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:35
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER height.
Resources	Processor Time	00:00:00.08
	Elapsed Time	00:00:00.09
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Height in inches[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.032[a]	.001	-.003	8.2808

a. Predictors: (Constant), Height in inches

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	17.322	1	17.322	.253	.616[b]
	Residual	17005.758	248	68.572		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Height in inches

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	26.264	14.119		1.860	.064
	Height in inches	-.101	.201	-.032	-.503	.616

a. Dependent Variable: Body Fat

```
2307  0
2308  0
2309  0  REGRESSION
2310  0    /MISSING LISTWISE
2311  0    /STATISTICS COEFF OUTS R ANOVA
2312  0    /CRITERIA=PIN(.05) POUT(.10)
```

```
2313  0    /NOORIGIN
2314  0    /DEPENDENT fat
2315  0    /METHOD=ENTER neck.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:35
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER neck.
Resources	Processor Time	00:00:00.13
	Elapsed Time	00:00:00.10
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Neck Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.490[a]	.240	.237	7.2241

a. Predictors: (Constant), Neck Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4080.533	1	4080.533	78.190	.000[b]
	Residual	12942.546	248	52.188		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Neck Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-44.301	7.193		-6.159	.000
	Neck Circumference	1.670	.189	.490	8.842	.000

a. Dependent Variable: Body Fat

```
2316  0
2317  0  REGRESSION
2318  0    /MISSING LISTWISE
2319  0    /STATISTICS COEFF OUTS R ANOVA
2320  0    /CRITERIA=PIN(.05) POUT(.10)
```

```
2321  0    /NOORIGIN
2322  0    /DEPENDENT fat
2323  0    /METHOD=ENTER chest.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:35
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER chest.
Resources	Processor Time	00:00:00.13
	Elapsed Time	00:00:00.06
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Chest Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.696[a]	.484	.482	5.9499

a. Predictors: (Constant), Chest Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	8243.625	1	8243.625	232.864	.000[b]
	Residual	8779.455	248	35.401		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Chest Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-50.375	4.573		-11.016	.000
	Chest Circumference	.689	.045	.696	15.260	.000

a. Dependent Variable: Body Fat

```
2324  0
2325  0  REGRESSION
2326  0    /MISSING LISTWISE
2327  0    /STATISTICS COEFF OUTS R ANOVA
2328  0    /CRITERIA=PIN(.05) POUT(.10)
```

```
2329  0    /NOORIGIN
2330  0    /DEPENDENT fat
2331  0    /METHOD=ENTER hip.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:35
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER hip.
Resources	Processor Time	00:00:00.08
	Elapsed Time	00:00:00.07
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Hip Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.613[a]	.375	.373	6.5475

a. Predictors: (Constant), Hip Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	6391.382	1	6391.382	149.088	.000[b]
	Residual	10631.697	248	42.870		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Hip Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-52.502	5.885		-8.922	.000
	Hip Circumference	.717	.059	.613	12.210	.000

a. Dependent Variable: Body Fat

```
2332  0
2333  0  REGRESSION
2334  0    /MISSING LISTWISE
2335  0    /STATISTICS COEFF OUTS R ANOVA
2336  0    /CRITERIA=PIN(.05) POUT(.10)
```

```
2337  0    /NOORIGIN
2338  0    /DEPENDENT fat
2339  0    /METHOD=ENTER thigh.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:35
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER thigh.
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.07
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Thigh Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.544[a]	.296	.293	6.9527

a. Predictors: (Constant), Thigh Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	5034.729	1	5034.729	104.152	.000[b]
	Residual	11988.351	248	48.340		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Thigh Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-32.545	5.087		-6.398	.000
	Thigh Circumference	.871	.085	.544	10.205	.000

a. Dependent Variable: Body Fat

```
2340  0
2341  0  REGRESSION
2342  0    /MISSING LISTWISE
2343  0    /STATISTICS COEFF OUTS R ANOVA
2344  0    /CRITERIA=PIN(.05) POUT(.10)
```

```
2345  0    /NOORIGIN
2346  0    /DEPENDENT fat
2347  0    /METHOD=ENTER knee.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:35
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER knee.
Resources	Processor Time	00:00:00.08
	Elapsed Time	00:00:00.09
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Knee Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.494[a]	.244	.241	7.2055

a. Predictors: (Constant), Knee Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4147.129	1	4147.129	79.877	.000[b]
	Residual	12875.951	248	51.919		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Knee Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-46.812	7.397		-6.329	.000
	Knee Circumference	1.710	.191	.494	8.937	.000

a. Dependent Variable: Body Fat

```
2348  0
2349  0  REGRESSION
2350  0    /MISSING LISTWISE
2351  0    /STATISTICS COEFF OUTS R ANOVA
2352  0    /CRITERIA=PIN(.05) POUT(.10)
```

```
2353  0    /NOORIGIN
2354  0    /DEPENDENT fat
2355  0    /METHOD=ENTER ankle.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:35
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER ankle.
Resources	Processor Time	00:00:00.11
	Elapsed Time	00:00:00.21
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Ankle Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.254[a]	.064	.061	8.0142

a. Predictors: (Constant), Ankle Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1094.854	1	1094.854	17.047	.000[b]
	Residual	15928.225	248	64.227		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Ankle Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-9.482	6.959		-1.363	.174
	Ankle Circumference	1.240	.300	.254	4.129	.000

a. Dependent Variable: Body Fat

```
2356  0
2357  0  REGRESSION
2358  0    /MISSING LISTWISE
2359  0    /STATISTICS COEFF OUTS R ANOVA
2360  0    /CRITERIA=PIN(.05) POUT(.10)
```

```
2361  0    /NOORIGIN
2362  0    /DEPENDENT fat
2363  0    /METHOD=ENTER biceps.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:36
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER biceps.
Resources	Processor Time	00:00:00.08
	Elapsed Time	00:00:00.08
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Extended Biceps Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.487[a]	.237	.234	7.2374

a. Predictors: (Constant), Extended Biceps Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4032.765	1	4032.765	76.990	.000[b]
	Residual	12990.314	248	52.380		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Extended Biceps Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-23.876	4.927		-4.846	.000
	Extended Biceps Circumference	1.333	.152	.487	8.774	.000

a. Dependent Variable: Body Fat

```
2364  0
2365  0  REGRESSION
2366  0    /MISSING LISTWISE
2367  0    /STATISTICS COEFF OUTS R ANOVA
```

```
2368  0    /CRITERIA=PIN(.05) POUT(.10)
2369  0    /NOORIGIN
2370  0    /DEPENDENT fat
2371  0    /METHOD=ENTER forearm.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:36
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER forearm.
Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.09
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Forearm Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.351[a]	.123	.120	7.7569

a. Predictors: (Constant), Forearm Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	2101.105	1	2101.105	34.920	.000[b]
	Residual	14921.974	248	60.169		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Forearm Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-22.228	7.023		-3.165	.002
	Forearm Circumference	1.444	.244	.351	5.909	.000

a. Dependent Variable: Body Fat

```
2372  0
2373  0  REGRESSION
2374  0    /MISSING LISTWISE
2375  0    /STATISTICS COEFF OUTS R ANOVA
2376  0    /CRITERIA=PIN(.05) POUT(.10)
```

```
2377  0    /NOORIGIN
2378  0    /DEPENDENT fat
2379  0    /METHOD=ENTER wrist.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:36
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER wrist.
Resources	Processor Time	00:00:00.09
	Elapsed Time	00:00:00.07
	Memory Required	5248 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Wrist Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.344[a]	.118	.115	7.7799

a. Predictors: (Constant), Wrist Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	2012.466	1	2012.466	33.249	.000[b]
	Residual	15010.614	248	60.527		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Wrist Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-36.624	9.689		-3.780	.000
	Wrist Circumference	3.059	.531	.344	5.766	.000

a. Dependent Variable: Body Fat

```
2380  0
2381  0
2382  0  *from above height with pvalue>0.2 is removed
2383  0
2384  0  *new model becomes;
```

```
2385  0
2386  0  REGRESSION
2387  0    /MISSING LISTWISE
2388  0    /STATISTICS COEFF OUTS R ANOVA
2389  0    /CRITERIA=PIN(.05) POUT(.10)
2390  0    /NOORIGIN
2391  0    /DEPENDENT fat
2392  0    /METHOD=ENTER age neck chest hip thigh knee ankle biceps forearm wrist.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:36
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age neck chest hip thigh knee ankle biceps forearm wrist.
Resources	Processor Time	00:00:00.13
	Elapsed Time	00:00:00.09

Notes

Memory Required	11648 bytes
Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Wrist Circumference, Age, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Hip Circumference[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.777[a]	.604	.588	5.3083

a. Predictors: (Constant), Wrist Circumference, Age, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Hip Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10288.450	10	1028.845	36.512	.000[b]
	Residual	6734.630	239	28.178		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Wrist Circumference, Age, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Hip Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-34.149	7.358		-4.641	.000
	Age	.236	.034	.362	6.968	.000
	Neck Circumference	-.407	.273	-.119	-1.490	.137
	Chest Circumference	.523	.088	.528	5.928	.000
	Hip Circumference	.252	.138	.215	1.833	.068
	Thigh Circumference	.508	.173	.317	2.935	.004
	Knee Circumference	-.235	.280	-.068	-.838	.403
	Ankle Circumference	-.045	.268	-.009	-.168	.867
	Extended Biceps Circumference	-.029	.207	-.011	-.141	.888
	Forearm Circumference	.334	.246	.081	1.358	.176
	Wrist Circumference	-2.659	.640	-.299	-4.157	.000

a. Dependent Variable: Body Fat

```
2393  0
2394  0  *eliminiate non significant variables using backward;
2395  0
2396  0  REGRESSION
2397  0    /MISSING LISTWISE
2398  0    /STATISTICS COEFF OUTS R ANOVA
2399  0    /CRITERIA=PIN(.05) POUT(.10)
2400  0    /NOORIGIN
2401  0    /DEPENDENT fat
2402  0    /METHOD=BACKWARD age neck chest hip thigh knee ankle biceps forearm wrist.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:36
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=BACKWARD age neck chest hip thigh knee ankle biceps forearm wrist.
Resources	Processor Time	00:00:00.13
	Elapsed Time	00:00:00.12
	Memory Required	12960 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Wrist Circumference, Age, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Hip Circumference[b]	.	Enter
2	.	Extended Biceps Circumference	Backward (criterion: Probability of F-to-remove >= .100).
3	.	Ankle Circumference	Backward (criterion: Probability of F-to-remove >= .100).
4	.	Knee Circumference	Backward (criterion: Probability of F-to-remove >= .100).
5	.	Forearm Circumference	Backward (criterion: Probability of F-to-remove >= .100).

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
6	.	Neck Circumference	Backward (criterion: Probability of F-to-remove >= .100).
7	.	Hip Circumference	Backward (criterion: Probability of F-to-remove >= .100).

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.777[a]	.604	.588	5.3083
2	.777[b]	.604	.590	5.2975
3	.777[c]	.604	.591	5.2868
4	.777[d]	.603	.591	5.2848
5	.775[e]	.600	.590	5.2928
6	.773[f]	.598	.589	5.2989
7	.771[g]	.594	.587	5.3129

a. Predictors: (Constant), Wrist Circumference, Age, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Hip Circumference

b. Predictors: (Constant), Wrist Circumference, Age, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Knee Circumference, Neck Circumference, Hip Circumference

c. Predictors: (Constant), Wrist Circumference, Age, Forearm Circumference, Thigh Circumference, Chest Circumference, Knee Circumference, Neck Circumference, Hip Circumference

d. Predictors: (Constant), Wrist Circumference, Age, Forearm Circumference, Thigh Circumference, Chest Circumference, Neck Circumference, Hip Circumference

e. Predictors: (Constant), Wrist Circumference, Age, Thigh Circumference, Chest Circumference, Neck Circumference, Hip Circumference

f. Predictors: (Constant), Wrist Circumference, Age, Thigh Circumference, Chest Circumference, Hip Circumference

g. Predictors: (Constant), Wrist Circumference, Age, Thigh Circumference, Chest Circumference

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10288.450	10	1028.845	36.512	.000[b]
	Residual	6734.630	239	28.178		
	Total	17023.080	249			
2	Regression	10287.888	9	1143.099	40.733	.000[c]
	Residual	6735.192	240	28.063		
	Total	17023.080	249			
3	Regression	10287.089	8	1285.886	46.006	.000[d]
	Residual	6735.991	241	27.950		
	Total	17023.080	249			
4	Regression	10264.156	7	1466.308	52.500	.000[e]
	Residual	6758.923	242	27.929		
	Total	17023.080	249			
5	Regression	10215.787	6	1702.631	60.779	.000[f]
	Residual	6807.292	243	28.014		
	Total	17023.080	249			
6	Regression	10171.906	5	2034.381	72.453	.000[g]
	Residual	6851.174	244	28.079		
	Total	17023.080	249			
7	Regression	10107.404	4	2526.851	89.518	.000[h]
	Residual	6915.675	245	28.227		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Wrist Circumference, Age, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Extended Biceps Circumference, Knee Circumference, Neck Circumference, Hip Circumference

c. Predictors: (Constant), Wrist Circumference, Age, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Knee Circumference, Neck Circumference, Hip Circumference

d. Predictors: (Constant), Wrist Circumference, Age, Forearm Circumference, Thigh Circumference, Chest

e. Predictors: (Constant), Wrist Circumference, Age, Forearm Circumference, Thigh Circumference, Chest Circumference, Neck Circumference, Hip Circumference

f. Predictors: (Constant), Wrist Circumference, Age, Thigh Circumference, Chest Circumference, Neck Circumference, Hip Circumference

g. Predictors: (Constant), Wrist Circumference, Age, Thigh Circumference, Chest Circumference, Hip Circumference

h. Predictors: (Constant), Wrist Circumference, Age, Thigh Circumference, Chest Circumference

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-34.149	7.358		-4.641	.000
	Age	.236	.034	.362	6.968	.000
	Neck Circumference	-.407	.273	-.119	-1.490	.137
	Chest Circumference	.523	.088	.528	5.928	.000
	Hip Circumference	.252	.138	.215	1.833	.068
	Thigh Circumference	.508	.173	.317	2.935	.004
	Knee Circumference	-.235	.280	-.068	-.838	.403
	Ankle Circumference	-.045	.268	-.009	-.168	.867
	Extended Biceps Circumference	-.029	.207	-.011	-.141	.888
	Forearm Circumference	.334	.246	.081	1.358	.176
	Wrist Circumference	-2.659	.640	-.299	-4.157	.000
2	(Constant)	-33.932	7.181		-4.725	.000
	Age	.236	.034	.362	6.983	.000
	Neck Circumference	-.412	.270	-.121	-1.525	.128
	Chest Circumference	.521	.087	.526	5.987	.000
	Hip Circumference	.253	.137	.216	1.840	.067
	Thigh Circumference	.501	.166	.313	3.021	.003
	Knee Circumference	-.234	.280	-.068	-.837	.403
	Ankle Circumference	-.045	.267	-.009	-.169	.866
	Forearm Circumference	.323	.233	.079	1.386	.167
	Wrist Circumference	-2.666	.636	-.300	-4.192	.000
3	(Constant)	-34.046	7.135		-4.772	.000
	Age	.238	.033	.363	7.188	.000

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
	Neck Circumference	-.409	.269	-.120	-1.521	.129
	Chest Circumference	.521	.087	.526	5.997	.000
	Hip Circumference	.252	.137	.215	1.839	.067
	Thigh Circumference	.502	.165	.313	3.030	.003
	Knee Circumference	-.246	.271	-.071	-.906	.366
	Forearm Circumference	.324	.233	.079	1.392	.165
	Wrist Circumference	-2.698	.607	-.303	-4.443	.000
4	(Constant)	-35.345	6.987		-5.059	.000
	Age	.234	.033	.358	7.133	.000
	Neck Circumference	-.391	.268	-.115	-1.459	.146
	Chest Circumference	.521	.087	.526	6.001	.000
	Hip Circumference	.221	.132	.188	1.665	.097
	Thigh Circumference	.460	.159	.288	2.894	.004
	Forearm Circumference	.305	.231	.074	1.316	.189
	Wrist Circumference	-2.841	.586	-.319	-4.849	.000
5	(Constant)	-33.467	6.850		-4.886	.000
	Age	.224	.032	.343	7.007	.000
	Neck Circumference	-.331	.265	-.097	-1.252	.212
	Chest Circumference	.542	.085	.547	6.346	.000
	Hip Circumference	.196	.131	.168	1.495	.136
	Thigh Circumference	.486	.158	.304	3.075	.002
	Wrist Circumference	-2.635	.565	-.296	-4.660	.000
6	(Constant)	-34.596	6.798		-5.089	.000
	Age	.223	.032	.341	6.970	.000
	Chest Circumference	.505	.080	.510	6.301	.000
	Hip Circumference	.199	.132	.170	1.516	.131
	Thigh Circumference	.449	.155	.280	2.886	.004
	Wrist Circumference	-2.946	.508	-.331	-5.796	.000
7	(Constant)	-32.290	6.643		-4.861	.000
	Age	.222	.032	.340	6.926	.000
	Chest Circumference	.564	.070	.569	8.036	.000
	Thigh Circumference	.611	.113	.381	5.398	.000
	Wrist Circumference	-2.834	.504	-.318	-5.620	.000

a. Dependent Variable: Body Fat

Excluded Variables[a]

Model		Beta In	t	Sig.	Partial Correlation	Collinearity Statistics Tolerance
2	Extended Biceps Circumference	-.011[b]	-.141	.888	-.009	.289
3	Extended Biceps Circumference	-.011[c]	-.142	.887	-.009	.289
	Ankle Circumference	-.009[c]	-.169	.866	-.011	.551
4	Extended Biceps Circumference	-.009[d]	-.122	.903	-.008	.289
	Ankle Circumference	-.020[d]	-.381	.704	-.025	.584
	Knee Circumference	-.071[d]	-.906	.366	-.058	.268
5	Extended Biceps Circumference	.021[e]	.292	.770	.019	.320
	Ankle Circumference	-.020[e]	-.375	.708	-.024	.584
	Knee Circumference	-.061[e]	-.782	.435	-.050	.270
	Forearm Circumference	.074[e]	1.316	.189	.084	.517
6	Extended Biceps Circumference	.004[f]	.061	.951	.004	.331
	Ankle Circumference	-.015[f]	-.284	.777	-.018	.587
	Knee Circumference	-.055[f]	-.706	.481	-.045	.271
	Forearm Circumference	.060[f]	1.081	.281	.069	.532
	Neck Circumference	-.097[f]	-1.252	.212	-.080	.274
7	Extended Biceps Circumference	-.003[g]	-.047	.963	-.003	.332
	Ankle Circumference	-.007[g]	-.132	.895	-.008	.593
	Knee Circumference	-.024[g]	-.318	.751	-.020	.288
	Forearm Circumference	.047[g]	.855	.393	.055	.543
	Neck Circumference	-.099[g]	-1.275	.203	-.081	.274
	Hip Circumference	.170[g]	1.516	.131	.097	.131

a. Dependent Variable: Body Fat

b. Predictors in the Model: (Constant), Wrist Circumference, Age, Ankle Circumference, Forearm Circumference, Thigh Circumference, Chest Circumference, Knee Circumference, Neck Circumference, Hip Circumference

c. Predictors in the Model: (Constant), Wrist Circumference, Age, Forearm Circumference, Thigh Circumference, Chest Circumference, Knee Circumference, Neck Circumference, Hip Circumference

d. Predictors in the Model: (Constant), Wrist Circumference, Age, Forearm Circumference, Thigh Circumference, Chest Circumference, Neck Circumference, Hip Circumference

f. Predictors in the Model: (Constant), Wrist Circumference, Age, Thigh Circumference, Chest Circumference, Hip Circumference

g. Predictors in the Model: (Constant), Wrist Circumference, Age, Thigh Circumference, Chest Circumference

```
2403  0
2404  0
2405  0  *now we form interaction terms from the signficant variables from above
2406  0
2407  0  COMPUTE age_chest  = age   * chest.
2408  0  COMPUTE age_thigh  = age   * thigh.
2409  0  COMPUTE age_wrist  = age   * wrist.
2410  0  COMPUTE chest_thigh = chest * thigh.
2411  0  COMPUTE chest_wrist = chest * wrist.
2412  0  COMPUTE thigh_wrist = thigh * wrist.
2413  0  EXECUTE.
2414  0
2415  0  *add to model and run backward
2416  0
2417  0
2418  0
2419  0  REGRESSION
2420  0    /MISSING LISTWISE
2421  0    /STATISTICS COEFF OUTS R ANOVA
2422  0    /CRITERIA=PIN(.05) POUT(.10)
2423  0    /NOORIGIN
2424  0    /DEPENDENT fat
2425  0    /METHOD=BACKWARD age chest thigh wrist age_chest age_thigh age_wrist chest_thi
gh chest_wrist
2426  0      thigh_wrist.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:36
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=BACKWARD age chest thigh wrist age_chest age_thigh age_wrist chest_thigh chest_wrist thigh_wrist.
Resources	Processor Time	00:00:00.14
	Elapsed Time	00:00:00.11
	Memory Required	12960 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	thigh_wrist, Age, Chest Circumference, Wrist Circumference, age_thigh, Thigh Circumference, age_chest, age_wrist, chest_thigh, chest_wrist[b]	.	Enter
2	.	age_chest	Backward (criterion: Probability of F-to-remove >= .100).
3	.	chest_wrist	Backward (criterion: Probability of F-to-remove >= .100).
4	.	Wrist Circumference	Backward (criterion: Probability of F-to-remove >= .100).
5	.	chest_thigh	Backward (criterion: Probability of F-to-remove >= .100).
6	.	age_thigh	Backward (criterion: Probability of F-to-remove >= .100).

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.790[a]	.624	.608	5.1764
2	.790[b]	.624	.610	5.1658
3	.790[c]	.624	.611	5.1563
4	.789[d]	.623	.612	5.1488
5	.789[e]	.623	.614	5.1385
6	.788[f]	.621	.613	5.1412

a. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, age_thigh, Thigh Circumference, age_chest, age_wrist, chest_thigh, chest_wrist

b. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, age_thigh, Thigh Circumference, age_wrist, chest_thigh, chest_wrist

c. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, age_thigh, Thigh Circumference, age_wrist, chest_thigh

d. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, age_thigh, Thigh Circumference, age_wrist, chest_thigh

e. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, age_thigh, Thigh Circumference, age_wrist

f. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Thigh Circumference, age_wrist

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10619.065	10	1061.906	39.631	.000[b]
	Residual	6404.015	239	26.795		
	Total	17023.080	249			
2	Regression	10618.454	9	1179.828	44.212	.000[c]
	Residual	6404.626	240	26.686		
	Total	17023.080	249			
3	Regression	10615.630	8	1326.954	49.910	.000[d]
	Residual	6407.449	241	26.587		
	Total	17023.080	249			
4	Regression	10607.745	7	1515.392	57.164	.000[e]
	Residual	6415.335	242	26.510		
	Total	17023.080	249			
5	Regression	10606.825	6	1767.804	66.951	.000[f]
	Residual	6416.254	243	26.404		
	Total	17023.080	249			
6	Regression	10573.680	5	2114.736	80.007	.000[g]
	Residual	6449.399	244	26.432		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, age_thigh, Thigh Circumference, age_chest, age_wrist, chest_thigh, chest_wrist

c. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, age_thigh, Thigh Circumference, age_wrist, chest_thigh, chest_wrist

d. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, age_thigh, Thigh Circumference, age_wrist, chest_thigh

e. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, age_thigh, Thigh Circumference, age_wrist, chest_thigh

f. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, age_thigh, Thigh Circumference, age_wrist

g. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Thigh Circumference, age_wrist

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-75.516	71.822		-1.051	.294
	Age	-1.126	.654	-1.723	-1.723	.086
	Chest Circumference	.453	1.451	.458	.312	.755
	Thigh Circumference	3.448	2.013	2.154	1.712	.088
	Wrist Circumference	-3.425	6.887	-.385	-.497	.619
	age_chest	.001	.005	.139	.151	.880
	age_thigh	-.008	.008	-.723	-1.028	.305
	age_wrist	.095	.043	2.821	2.217	.028
	chest_thigh	-.005	.010	-.617	-.504	.615
	chest_wrist	.021	.071	.589	.301	.764
	thigh_wrist	-.102	.140	-1.670	-.727	.468
2	(Constant)	-74.172	71.123		-1.043	.298
	Age	-1.129	.652	-1.728	-1.732	.085
	Chest Circumference	.486	1.432	.490	.339	.735
	Thigh Circumference	3.440	2.009	2.149	1.713	.088
	Wrist Circumference	-3.725	6.582	-.419	-.566	.572
	age_thigh	-.008	.007	-.666	-1.128	.261
	age_wrist	.098	.039	2.897	2.487	.014
	chest_thigh	-.006	.010	-.656	-.549	.583
	chest_wrist	.023	.070	.629	.325	.745
	thigh_wrist	-.101	.140	-1.660	-.725	.469
3	(Constant)	-77.875	70.076		-1.111	.268
	Age	-1.240	.554	-1.898	-2.238	.026
	Chest Circumference	.907	.611	.916	1.485	.139
	Thigh Circumference	2.879	1.029	1.799	2.799	.006
	Wrist Circumference	-3.568	6.552	-.401	-.545	.587
	age_thigh	-.008	.007	-.671	-1.139	.256
	age_wrist	.104	.034	3.085	3.056	.002
	chest_thigh	-.006	.010	-.669	-.562	.575
	thigh_wrist	-.070	.100	-1.143	-.695	.488
4	(Constant)	-111.573	32.843		-3.397	.001
	Age	-1.146	.526	-1.753	-2.180	.030
	Chest Circumference	.625	.323	.631	1.933	.054
	Thigh Circumference	3.322	.630	2.075	5.270	.000

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
	age_thigh	-.007	.006	-.585	-1.033	.303
	age_wrist	.096	.030	2.837	3.155	.002
	chest_thigh	-.001	.005	-.115	-.186	.852
	thigh_wrist	-.123	.025	-2.012	-4.937	.000
5	(Constant)	-106.363	17.165		-6.197	.000
	Age	-1.175	.501	-1.798	-2.345	.020
	Chest Circumference	.566	.069	.572	8.170	.000
	Thigh Circumference	3.275	.576	2.046	5.690	.000
	age_thigh	-.007	.006	-.613	-1.120	.264
	age_wrist	.098	.027	2.913	3.646	.000
	thigh_wrist	-.125	.022	-2.047	-5.701	.000
6	(Constant)	-88.395	6.123		-14.437	.000
	Age	-1.386	.465	-2.121	-2.984	.003
	Chest Circumference	.553	.068	.558	8.093	.000
	Thigh Circumference	2.836	.422	1.771	6.722	.000
	age_wrist	.088	.025	2.602	3.471	.001
	thigh_wrist	-.116	.021	-1.909	-5.657	.000

a. Dependent Variable: Body Fat

Excluded Variables[a]

Model		Beta In	t	Sig.	Partial Correlation	Collinearity Statistics Tolerance
2	age_chest	.139[b]	.151	.880	.010	.002
3	age_chest	.177[c]	.194	.846	.013	.002
	chest_wrist	.629[c]	.325	.745	.021	.000
4	age_chest	.295[d]	.338	.736	.022	.002
	chest_wrist	.549[d]	.285	.776	.018	.000
	Wrist Circumference	-.401[d]	-.545	.587	-.035	.003
5	age_chest	.286[e]	.328	.743	.021	.002
	chest_wrist	.595[e]	.315	.753	.020	.000
	Wrist Circumference	-.048[e]	-.125	.901	-.008	.011
	chest_thigh	-.115[e]	-.186	.852	-.012	.004
6	age_chest	-.284[f]	-.407	.685	-.026	.003
	chest_wrist	.793[f]	.421	.674	.027	.000
	Wrist Circumference	.099[f]	.277	.782	.018	.012
	chest_thigh	-.280[f]	-.468	.640	-.030	.004
	age_thigh	-.613[f]	-1.120	.264	-.072	.005

a. Dependent Variable: Body Fat

b. Predictors in the Model: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, age_thigh, Thigh Circumference, age_wrist, chest_thigh, chest_wrist

c. Predictors in the Model: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, age_thigh, Thigh Circumference, age_wrist, chest_thigh

d. Predictors in the Model: (Constant), thigh_wrist, Age, Chest Circumference, age_thigh, Thigh Circumference, age_wrist, chest_thigh

e. Predictors in the Model: (Constant), thigh_wrist, Age, Chest Circumference, age_thigh, Thigh Circumference, age_wrist

f. Predictors in the Model: (Constant), thigh_wrist, Age, Chest Circumference, Thigh Circumference

```
2427  0  *in the above model; interaction terms of; age_wrist and thigh_wrist are retaine
d; but wrist was dropped; so we first bring
            it back
2428  0
2429  0  REGRESSION
2430  0    /MISSING LISTWISE
2431  0    /STATISTICS COEFF OUTS R ANOVA
2432  0    /CRITERIA=PIN(.05) POUT(.10)
2433  0    /NOORIGIN
2434  0    /DEPENDENT fat
```

```
2435  0    /METHOD=ENTER age chest thigh wrist age_wrist thigh_wrist.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:36
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age chest thigh wrist age_wrist thigh_wrist.
Resources	Processor Time	00:00:00.08
	Elapsed Time	00:00:00.08
	Memory Required	8160 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	thigh_wrist, Age, Chest Circumference, Wrist Circumference, Thigh Circumference, age_wrist[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.788[a]	.621	.612	5.1510

a. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, Thigh Circumference, age_wrist

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10575.712	6	1762.619	66.433	.000[b]
	Residual	6447.367	243	26.532		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, Thigh Circumference, age_wrist

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-104.489	58.480		-1.787	.075
	Age	-1.310	.541	-2.004	-2.423	.016
	Chest Circumference	.554	.069	.560	8.078	.000
	Thigh Circumference	3.040	.850	1.899	3.577	.000
	Wrist Circumference	.883	3.191	.099	.277	.782
	age_wrist	.083	.030	2.477	2.827	.005
	thigh_wrist	-.128	.046	-2.094	-2.800	.006

a. Dependent Variable: Body Fat

```
2436  0  *all other variables remain signficant; so we keep wrist;
2437  0
2438  0
2439  0  *check for confounding
2440  0  *crude model is as follows;
2441  0
2442  0  REGRESSION
2443  0    /MISSING LISTWISE
2444  0    /STATISTICS COEFF OUTS R ANOVA
2445  0    /CRITERIA=PIN(.05) POUT(.10)
2446  0    /NOORIGIN
2447  0    /DEPENDENT fat
2448  0    /METHOD=ENTER age chest thigh wrist age_wrist thigh_wrist
```

Regression

Notes

Output Created		03-NOV-2026 15:38:36
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age chest thigh wrist age_wrist thigh_wrist
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.08
	Memory Required	8160 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	thigh_wrist, Age, Chest Circumference, Wrist Circumference, Thigh Circumference, age_wrist[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.788[a]	.621	.612	5.1510

a. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, Thigh Circumference, age_wrist

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10575.712	6	1762.619	66.433	.000[b]
	Residual	6447.367	243	26.532		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), thigh_wrist, Age, Chest Circumference, Wrist Circumference, Thigh Circumference, age_wrist

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-104.489	58.480		-1.787	.075
	Age	-1.310	.541	-2.004	-2.423	.016
	Chest Circumference	.554	.069	.560	8.078	.000
	Thigh Circumference	3.040	.850	1.899	3.577	.000
	Wrist Circumference	.883	3.191	.099	.277	.782
	age_wrist	.083	.030	2.477	2.827	.005
	thigh_wrist	-.128	.046	-2.094	-2.800	.006

a. Dependent Variable: Body Fat

```
2449  0
2450  0  *the last variables to be removed follow this order; hip,neck,forearm,knee,ankle
 and biceps was the first to be removed
2451  0      *so we shall add hip and then compare the model with crude; and so on and so
 forth
2452  0
2453  0
2454  0  *adjusted model for hip below
2455  0
2456  0  DATASET ACTIVATE DataSet1.
2457  0  REGRESSION
2458  0    /MISSING LISTWISE
2459  0    /STATISTICS COEFF OUTS R ANOVA
2460  0    /CRITERIA=PIN(.05) POUT(.10)
2461  0    /NOORIGIN
2462  0    /DEPENDENT fat
2463  0    /METHOD=ENTER age chest thigh wrist age_wrist thigh_wrist hip.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:37
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age chest thigh wrist age_wrist thigh_wrist hip.
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.06
	Memory Required	8944 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Hip Circumference, Age, Wrist Circumference, Chest Circumference, Thigh Circumference, thigh_wrist, age_wrist[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.793[a]	.629	.619	5.1067

a. Predictors: (Constant), Hip Circumference, Age, Wrist Circumference, Chest Circumference, Thigh Circumference, thigh_wrist, age_wrist

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10712.025	7	1530.289	58.680	.000[b]
	Residual	6311.055	242	26.079		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Hip Circumference, Age, Wrist Circumference, Chest Circumference, Thigh Circumference, thigh_wrist, age_wrist

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-145.585	60.701		-2.398	.017
	Age	-1.208	.538	-1.848	-2.246	.026
	Chest Circumference	.469	.078	.474	6.045	.000
	Thigh Circumference	3.344	.853	2.089	3.920	.000
	Wrist Circumference	2.743	3.266	.308	.840	.402
	age_wrist	.078	.029	2.311	2.652	.009
	thigh_wrist	-.158	.047	-2.584	-3.348	.001
	Hip Circumference	.302	.132	.258	2.286	.023

a. Dependent Variable: Body Fat

```
2464  0  *Chest, wrist and thighwrist have %change >10%, so hip is a confounder
2465  0
2466  0  *adjusted model for neck
2467  0
2468  0  REGRESSION
2469  0    /MISSING LISTWISE
2470  0    /STATISTICS COEFF OUTS R ANOVA
2471  0    /CRITERIA=PIN(.05) POUT(.10)
2472  0    /NOORIGIN
2473  0    /DEPENDENT fat
2474  0    /METHOD=ENTER age chest thigh wrist hip age_wrist thigh_wrist neck.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:37
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age chest thigh wrist hip age_wrist thigh_wrist neck.
Resources	Processor Time	00:00:00.08
	Elapsed Time	00:00:00.06
	Memory Required	9776 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Neck Circumference, Age, Wrist Circumference, Thigh Circumference, Chest Circumference, Hip Circumference, thigh_wrist, age_wrist[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.794[a]	.630	.618	5.1119

a. Predictors: (Constant), Neck Circumference, Age, Wrist Circumference, Thigh Circumference, Chest Circumference, Hip Circumference, thigh_wrist, age_wrist

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10725.427	8	1340.678	51.305	.000[b]
	Residual	6297.652	241	26.131		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Neck Circumference, Age, Wrist Circumference, Thigh Circumference, Chest Circumference, Hip Circumference, thigh_wrist, age_wrist

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-141.194	61.070		-2.312	.022
	Age	-1.190	.539	-1.820	-2.207	.028
	Chest Circumference	.491	.083	.495	5.888	.000
	Thigh Circumference	3.291	.857	2.056	3.840	.000
	Wrist Circumference	2.720	3.270	.306	.832	.406
	Hip Circumference	.298	.132	.254	2.247	.026
	age_wrist	.077	.029	2.283	2.614	.010
	thigh_wrist	-.153	.047	-2.517	-3.234	.001
	Neck Circumference	-.185	.258	-.054	-.716	.475

a. Dependent Variable: Body Fat

```
2475  0
2476  0  *adjusted forearm
2477  0
2478  0  REGRESSION
2479  0    /MISSING LISTWISE
2480  0    /STATISTICS COEFF OUTS R ANOVA
2481  0    /CRITERIA=PIN(.05) POUT(.10)
2482  0    /NOORIGIN
2483  0    /DEPENDENT fat
2484  0    /METHOD=ENTER age chest thigh wrist hip age_wrist thigh_wrist forearm.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:37
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age chest thigh wrist hip age_wrist thigh_wrist forearm.
Resources	Processor Time	00:00:00.09
	Elapsed Time	00:00:00.07
	Memory Required	9776 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Forearm Circumference, age_wrist, Hip Circumference, Wrist Circumference, Chest Circumference, Thigh Circumference, thigh_wrist, Age[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.793[a]	.629	.617	5.1164

a. Predictors: (Constant), Forearm Circumference, age_wrist, Hip Circumference, Wrist Circumference, Chest Circumference, Thigh Circumference, thigh_wrist, Age

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10714.333	8	1339.292	51.162	.000[b]
	Residual	6308.747	241	26.177		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Forearm Circumference, age_wrist, Hip Circumference, Wrist Circumference, Chest Circumference, Thigh Circumference, thigh_wrist, Age

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-141.782	62.149		-2.281	.023
	Age	-1.211	.539	-1.853	-2.247	.026
	Chest Circumference	.463	.080	.468	5.778	.000
	Thigh Circumference	3.272	.888	2.044	3.684	.000
	Wrist Circumference	2.458	3.410	.276	.721	.472
	Hip Circumference	.305	.133	.260	2.297	.022
	age_wrist	.078	.029	2.320	2.655	.008
	thigh_wrist	-.154	.049	-2.525	-3.161	.002
	Forearm Circumference	.068	.228	.016	.297	.767

a. Dependent Variable: Body Fat

```
2485  0
2486  0  *knee adjusted
2487  0
2488  0  REGRESSION
2489  0    /MISSING LISTWISE
2490  0    /STATISTICS COEFF OUTS R ANOVA
2491  0    /CRITERIA=PIN(.05) POUT(.10)
2492  0    /NOORIGIN
2493  0    /DEPENDENT fat
2494  0    /METHOD=ENTER age chest thigh wrist hip age_wrist thigh_wrist forearm knee.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:37
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age chest thigh wrist hip age_wrist thigh_wrist forearm knee.
Resources	Processor Time	00:00:00.11
	Elapsed Time	00:00:00.08
	Memory Required	10688 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Knee Circumference, Age, Forearm Circumference, Wrist Circumference, Chest Circumference, Thigh Circumference, Hip Circumference, thigh_wrist, age_wrist[b]	.	Enter

a. Dependent Variable: Body Fat

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.797[a]	.635	.621	5.0877

a. Predictors: (Constant), Knee Circumference, Age, Forearm Circumference, Wrist Circumference, Chest Circumference, Thigh Circumference, Hip Circumference, thigh_wrist, age_wrist

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10810.724	9	1201.192	46.405	.000[b]
	Residual	6212.355	240	25.885		
	Total	17023.080	249			

a. Dependent Variable: Body Fat

b. Predictors: (Constant), Knee Circumference, Age, Forearm Circumference, Wrist Circumference, Chest Circumference, Thigh Circumference, Hip Circumference, thigh_wrist, age_wrist

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-149.656	61.935		-2.416	.016
	Age	-1.361	.542	-2.082	-2.513	.013
	Chest Circumference	.457	.080	.462	5.730	.000
	Thigh Circumference	3.649	.905	2.280	4.034	.000
	Wrist Circumference	3.289	3.418	.370	.962	.337
	Hip Circumference	.379	.137	.324	2.758	.006
	age_wrist	.087	.030	2.573	2.928	.004
	thigh_wrist	-.170	.049	-2.785	-3.457	.001
	Forearm Circumference	.084	.227	.020	.369	.712
	Knee Circumference	-.517	.268	-.149	-1.930	.055

a. Dependent Variable: Body Fat

```
2495  0
2496  0  *ankle adjusted
2497  0
2498  0  REGRESSION
2499  0    /MISSING LISTWISE
2500  0    /STATISTICS COEFF OUTS R ANOVA
2501  0    /CRITERIA=PIN(.05) POUT(.10)
2502  0    /NOORIGIN
2503  0    /DEPENDENT fat
2504  0    /METHOD=ENTER age chest thigh wrist hip age_wrist thigh_wrist forearm knee ank
le.
```

Regression

Notes

Output Created		03-NOV-2026 15:38:37
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\body fat final.sav
	Active Dataset	DataSet1
	Filter	fat > 0 & height > 30 (FILTER)
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	250
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT fat /METHOD=ENTER age chest thigh wrist hip age_wrist thigh_wrist forearm knee ankle.
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.09
	Memory Required	11648 bytes
	Additional Memory Required for Residual Plots	0 bytes